MIZORAM PUBLIC SERVICE COMMISSION

TECHNICAL COMPETITIVE EXAMINATION FOR JUNIOR GRADE OF MIZORAM HEALTH SERVICES (COMBINED) (GDMO SUB-CADRE) UNDER HEALTH & FAMILY WELFARE DEPARTMENT, GOVERNMENT OF MIZORAM. (SEPTEMBER-2025)

TECHNICAL SUBJECT PAPER-III (ALLOPATHY)

Time Allowed : 2 hours FM : 200

All questions carry equal marks of 2 each.
Attempt all questions.

1. In a village of 2,000 people, 10 new cases of a disease were detected over one year. If the average duration of the disease is 2 years, the expected prevalence (at steady state) is approximately:
(a) 10/4000 per 1000 population (b) 20/2000 per 1000 population
(c) 10% (d) 0.5%

2. Virulence of a disease is indicated by:
(a) Proportional mortality rate (b) Specific mortality rate
(c) Case fatality rate (d) Amount of GDP spent on control of disease

3. Natural history of disease is best studied by:
(a) Cross sectional study (b) RCT
(c) Case-control study (d) Cohort study

4. An epidemic of Hepatitis A is an example of:
(a) Common source, single exposure epidemic
(b) Common source, continuous exposure epidemic
(c) Propagated epidemic
(d) Slow epidemic

5. Best indicators for air pollution are: Indicators for Air Quality Index for monitoring of Air pollution Include:
(a) Sulphur dioxide, smoke, and Suspended particles
(b) Sulphur dioxide, hydrogen sulphide, carbon monoxide
(c) Carbon dioxide, hydrogen sulphide, lead
(d) Sulphur dioxide, lead and particulate matter

6. The test used to identify frozen vaccine is:
(a) Shake test (b) Habel test
(c) Shick test (d) Heinz test

7. A study is to be designed to understand the correct technique for injection by a health care professional. The best way to assess will be:
(a) In depth interview with the patient
(b) Using CCTV
(c) In depth interview with the health care professional
(d) Questionnaire based with checklist

8. Any restriction or lack of ability to perform an activity within the range that is considered normal for a human being is called:
(a) Disease (b) Impairment
(c) Disability (d) Handicap

9. Plant source with highest protein:
(a) Black gram (b) Dry peas
(c) Soybean (d) Bengal gram

10. Extra calories per day in lactating mothers in first six months:
(a) 300 (b) 500
(c) 600 (d) 1000

11. Which of the following formulates food standards for inter- national market:
(a) AGMARK (b) ISI
(c) PFA (d) Codex alimentarius

12. As per NPCB, cut off for blindness is a vision of:
(a) <3/60 in worse eye (b) <6/60 in better eye
(c) <3/60 in better eye (d) <6/60 in worse eye

13. Caisson disease is associated with:
(a) Mountain climbing (b) Deep sea diving
(c) Flying (d) Space travel

14. All of the following have a common vector except:
(a) Dengue fever (b) Yellow fever
(c) Kala Azar (d) Japanese encephalitis

15. Most common cause of maternal death in India is:
(a) Infection (b) Eclampsia
(c) Anemia (d) Bleeding

16. The most serious complication of influenza is:
(a) Myocarditis (b) Pneumonia
(c) Encephalitis (d) Pancreatitis

17. WHO defines adolescent age between:
(a) 10-19 years of age (b) 10-14 years of age
(c) 10-25 years of age (d) 9-14 years of age

18. Mechanism of action of Intra Uterine Device is all except:
(a) Foreign body reaction (b) Thickening of cervical mucus
(c) Unfavorable endometrium for implantation (d) Thinning of fallopian tube

19. Triage is:
(a) Treating the most serious cases.
(b) Treating mentally ill patients
(c) Categorization of patients and treating them according to the available resources
(d) Treating terminally ill patients

20. '*Critical path*' in network analysis is:
(a) Most expensive path in a network
(b) Congested path in a network
(c) Shortest path in a network
(d) Longest path in a network

21. Dengue hemorrhagic fever discharge criteria is:
(a) After 24 hours fever controlled on paracetamol
(b) After return of appetite
(c) Urine output 200ml/day
(d) 24 hrs after control of shock

22. A patient with sputum positive pulmonary TB who has taken anti-TB drugs for <4 weeks is classified as:
(a) Relapse case
(b) Default case
(c) Treatment failure case
(d) New case

23. HIV postexposure prophylaxis should be started in:
(a) 2 hours
(b) 24 hours
(c) 48 hours
(d) 72 Hours

24. DASH is the acronym for:
(a) Domestic approach to safeguard hepatitis
(b) Dietary approach to stop hyperlipidemia
(c) Dietary approach to stop hypertension
(d) Domestic approach to stop hypertension

25. Which of the following vaccines is most sensitive to heat?
(a) Oral polio vaccine
(b) BCG vaccine
(c) DPT vaccine
(d) Measles vaccine

26. All are processes that influence demography except:
(a) Mortality
(b) Fertility
(c) Migration
(d) Life expectancy

27. Under the Global Strategy for Women's, Children's and Adolescent's Health 2016 - 2030, the goal for maternal mortality reduction is:
(a) less than 70 per 1,00,000 live births
(b) less than 75 per 1,00,000 live births
(c) less than 80 per 1,00,000 live births
(d) less than 85 per 1,00,000 live births

28. The sex-ratio at birth in India as per NFHS-5 is:
(a) 985
(b) 990
(c) 1010
(d) 1020

29. Mode of transmission of Zika virus are all except:
(a) Sharing of needles
(b) Sexual contact
(c) Blood transfusion
(d) Organ transplantation

30. Open Vial Policy is applicable to all the following vaccines except:
(a) OPV
(b) Pentavalent vaccine
(c) Rota virus vaccine
(d) Pneumococcal Conjugate vaccine

31. Reverse cold chain is a strategy used for the surveillance of which disease?
(a) Poliomyelitis
(b) Measles
(c) COVID-19
(d) Influenza

32. Ability of a test to correctly identify those who do not have a disease is called?
(a) Sensitivity
(b) Specificity
(c) Accuracy
(d) Negative Predictive Value

33. As per the Indian Public Health Standards for Health and Wellness Centre - Primary Health Centre, the population norm for urban areas is-

(a) 20,000 (b) 30,000
(c) 40,000 (d) 50,000

34. All the following parameters are taken into account for classification of Socio-economic classes as per Kuppuswamy scale except:

(a) Education status (b) Occupational status
(c) Per capita Income (d) Gross Domestic Product

35. For purification of water by chlorination the minimum recommended concentration of free chlorine should be:

(a) 0.1 mg/L per hour (b) 0.01mg/L per hour
(c) 0.5 mg/L per hour (d) 0.05 mg/L per hour

36. Which species of mosquito breeds in dirty water?

(a) Anopheles (b) Aedes
(c) Mansonia (d) Culex

37. Under the Employee State Insurance Act, the maximum amount payable for funeral expenses is:

(a) Rs 10,000 (b) Rs 15,000
(c) Rs 20,000 (d) Rs 25,000

38. What is the minimum population of an urban area to be organized as a Corporation?

(a) < 2.5 lakhs (b) <2 lakhs
(c) >2 lakhs (d) >2.5 lakhs

39. A public health researcher is investigating an outbreak of a rare infectious disease in a community. The disease is characterized by high fever, headache, joint pain, and a rash. The researcher collects data on cases and non-cases, exposure history, and demographic information. Which type of study design is most suitable for this outbreak investigation?

(a) Cross-sectional study (b) Cohort study
(c) Case-control study (d) Ecological study

40. Screening of a disease is an example of what level of prevention?

(a) Primordial (b) Primary
(c) Secondary (d) Tertiary

41. What measurement scale is Likert Scale?

(a) Nominal (b) Ordinal
(c) Interval (d) Ratio

42. What social problem does the campaign '*Nasha Mukt Bharat Abhiyaan*' address?

(a) Child trafficking (b) Domestic violence
(c) Alcohol abuse (d) Substance abuse

43. The ability of a measurement process to correctly measure what it actually tends to measure is known as:

(a) Reliability (b) Reproducibility
(c) Validity (d) Consistency

44. The continuous oversight of activities to ensure that they are proceeding according to plan is known as:

(a) Surveillance (b) Evaluation
(c) Monitoring (d) Assessment

45. When a disease agent is permitted to persist in a community at a level where it ceases to be a public health problem, the phase is called:
(a) Control (b) Elimination
(c) Eradication (d) Low prevalence

46. Which day is celebrated as '*World Mental Health Day*'?
(a) 4th February (b) 7th April
(c) 10th October (d) 1st December

47. In immunization, Zero Dose (ZD) children is used for referring those who have not received a first dose of which vaccine by one year of age?
(a) OPV (b) Pentavalent
(c) BCG (d) Hepatitis B

48. What is the tool for control of confounding in an experimental study design?
(a) Blinding (b) Statistical analysis
(c) Randomization (d) None of the above

49. Use of Combined Oral Contraceptive accounts for an increase in the risk of:
(a) Breast cancer (b) Ovarian cancer
(c) Endometrial cancer (d) None of the above

50. In India, the law pertaining to prevention of RTAs are laid down in which of the following legislation?
(a) Road Traffic Authority Act (b) Motor Vehicles Act
(c) Traffic Violence Act (d) Prevention of Immoral Traffic Act

51. Asphyxia denotes to a condition in which there is a deprivation of :
(a) Nitrogen (b) Oxygen
(c) Hydrogen (d) Carbon Dioxide .

52. The classic post-mortem finding in case of death due to drowning is:
(a) Presence of frothy fluid in the airways (b) Fracture of the cervical spine
(c) Ruptured heart and lungs (d) Abdominal injuries due to blunt force

53. Patterned abrasion is a variety of:
(a) Linear abrasion (b) Pressure abrasion
(c) Sliding abrasion (d) Superficial bruise

54. The most common cause of removing name of a doctor from the Medical Register is:
(a) Fraud cases (b) Death of the doctor
(c) Wrong error (d) Serious professional misconduct

55. BNS section for issuing a false certificate:
(a) 231 BNS (b) 232 BNS
(c) 233 BNS (d) 234 BNS

56. Filigree burns occur in:
(a) Lightening (b) Electrocution
(c) Vitriolage (d) Infanticide

57. The characteristic smell of bitter almonds is associated with poisoning by:
(a) Cyanide (b) Mercury
(c) Arsenic (d) Carbon monoxide

58. Davidson body is found in:
(a) Neutrophil leucocytes (b) Monocytes
(c) T-lymphocytes (d) Buccal smears

59. The study of hair is:
(a) Trichology (b) Cheiloscopy
(c) Virtopsy (d) None of the above

60. Post-mortem staining begins to appear:
(a) Immediately after death (b) Within 6-12 hours after death
(c) Within 24 hours after death (d) Within 48 hours after death

61. Time limit for exhumation in India is:
(a) 7 years (b) 10 years
(c) 25 years (d) No time limit

62. Hyoid fracture is common in:
(a) Road traffic accidents (b) Hanging
(c) Throttling (d) Choking

63. Irresistible sexual desire in males is known as:
(a) Sadism (b) Tribadism
(c) Nymphomania (d) Satyriasis

64. Deliriant poison is:
(a) Datura (b) Opium
(c) Methyl alcohol (d) Barbiturate

65. Cerebral cortical cells being more sensitive to circulatory arrest begin to die within :
(a) 5 minutes (b) 3 minutes
(c) 10 minutes (d) 15 minutes

66. In India how many inquest is/are there :
(a) 3 (b) 5
(c) 1 (d) 2

67. To determine Sex in the skeletal remains, the most important bones is:
(a) Femur (b) Tibia
(c) Pelvis (d) Humerus

68. Which section of the new law BNS describes '*Rape*' ?
(a) Section 50 (b) Section 75
(c) Section 38 (d) Section 63

69. When alcohols enter human body absorption starts from?
(a) Oesophagus (b) Mouth & Tongue
(c) Stomach & small intestine (d) Large intestine

70. Cherry red coloration of the skin and mucus membrane is a signs of:
(a) CO poisoning (b) CO_2 poisoning
(c) H_2S poisoning (d) Snake bite

71. In case of burns, medical officers can determine if burns are ante-mortem by :
(a) cyanosis of fingernails (b) congestions of organs
(c) pale internal organs (d) soot in the respiratory passages

72. Spurting and clotting of blood takes place only in:
(a) ante-mortem wound (b) post-mortem wound
(c) poisoning (d) electrocution

73. A doctor should not take fees for issuing certificate:
(a) Death certificate (b) To a govt. employee
(c) Poor patient (d) Psychiatric patient

74. Mechanical injury, which have length and breadth but no depth is called :
(a) Abrasion (b) Laceration
(c) Contusion (d) Incised wound

75. Medical officer in the witness box is:
(a) An expert witness (b) Common witness
(c) Both expert and common witness (d) None of the above

Direction (Question Nos. 76 - 79) : Look at the word in capital letters. Then, from the four words that follow, find the one that means the same thing as the word in capitals.

76. WHIMSICAL
(a) unlike (b) musical
(c) dancing (d) unpredictable

77. APATHETIC
(a) ignorant (b) indifferent
(c) pitiful (d) concerned

78. CLANDESTINE
(a) furtive (b) safe
(c) tribal (d) open

79. INCULCATE
(a) grow (b) inquire
(c) instill (d) compute

80. First read the saying or proverb in capital letters. Then, from the five sayings under it, find the one that means the same thing, and mark your answer.

LET ANOTHER'S SHIPWRECK BE YOUR GUIDE AT SEA
(a) one man's meat is another man's poison
(b) ships fear fire more than water
(c) wise men learn from other men's mistakes
(d) the worst things happen at sea

81. Please select a pair of words that have similar relationships

Flower – bouquet
(a) soldier – company (b) bread – cookie
(c) magazine – cover (d) tap – water

82. Find the odd one out from the following anagrams:
(a) SCHAMOT (b) LABLOTOF
(c) CEKTIRC (d) SNINET

Direction (Question No. 83) : Pick out the correct synonym of the word underlined.

83. The police officer found that the documents recovered were <u>fabricated.</u>

(a) constricted
(b) genuine
(c) forged
(d) prepared

Directions (Question No. 84) : Find the Missing number in the series given below.

84. 121, 112, ?, 97, 91, 86

(a) 102
(b) 108
(c) 99
(d) 104

85. Among four playing cards, a Heart is to the immediate left of a spade, a 9 is to the left of a J, a Diamond to the right of a Heart, a 7 is the left most card and a Club is the right most card. An 8 is to the left of a 9. Then the 9 is a _____.

(a) Spade
(b) Heart
(c) Diamond
(d) Club

Direction (Question Nos. 86 & 87) : Given below are four sets of letters. Three of the four sets follow a certain rule. One set does not. Mark the letter of the one set that does not follow the rule.

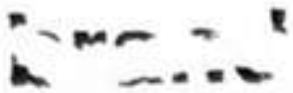
86.

(a) CDED
(b) LMOM
(c) PQRQ
(d) STUT

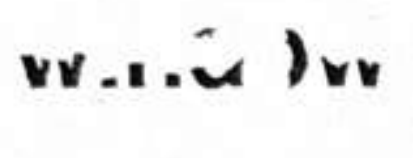
87.

(a) DECG
(b) JKIL
(c) MNLO
(d) QRPS

Direction (Question Nos. 88 & 89) : In this test, you will see a word with parts of the letters missing. This "incomplete" word is followed by four "jumbled" words. You are first to use your imagination to figure out what the incomplete word is. Then find, among the four scrambled choices, the one that has the right letters to spell the incomplete word.

88. 

(a) rodba
(b) dnbra
(c) aibnr
(d) radeb

89. 

(a) lowilw
(b) rdweno
(c) dwowni
(d) nawred

90. The average height of students in a class of 10 is 105cm. If 20 more students with an average height of 120cm join the class, what will the average height be?

(a) 105 cm (b) 110 cm
(c) 112 cm (d) 115 cm

91. A student has to secure 35% marks to pass. He got 80 marks and failed by 60 marks. The maximum marks are:

(a) 100 (b) 200
(c) 300 (d) 400

92. 125% of 32 = _____

(a) 6 (b) 12
(c) 36 (d) 40

93. Which has the longest perimeter?

(a) a square of side 10 cms
(b) a rectangle of sides 12 cms and 9 cms
(c) a circle of radius 7 cms
(d) a rhombus of side 9 cms

94. $\frac{5}{7}$ of 245 is -

(a) 165 (b) 175
(c) 207 (d) 343

95. Which one of the numbers below could replace X in both places in 4 + X = 6 - X?

(a) 4 (b) 3
(c) 2 (d) 1

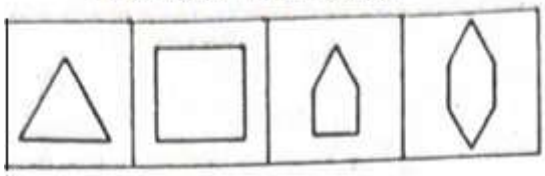
96. The third smallest 3-digit odd number using the digits 1 to 9 is ___

(a) 105 (b) 123
(c) 113 (d) 115

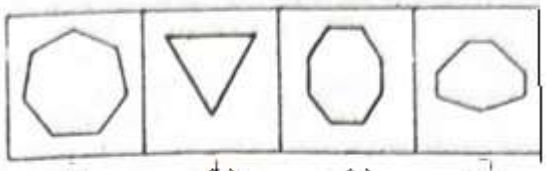
Direction (Question Nos. 97 - 99) : In this test you will see rows of designs or figures. Each row has four designs called Problem figures. These are followed by four designs called Answer figures. The four Problem figures make a series. That means, they follow each other in a special order. The fifth figure is missing. Find the right picture among the four Answer figures and mark the correct one.

97.

PROBLEM FIGURES



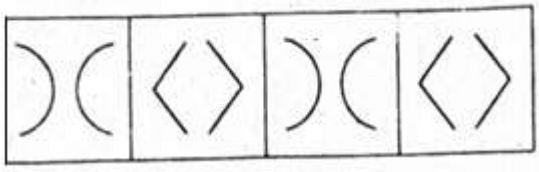
ANSWER FIGURES



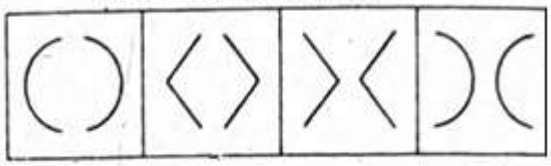
(a) (b) (c) (d)

98.

PROBLEM FIGURES



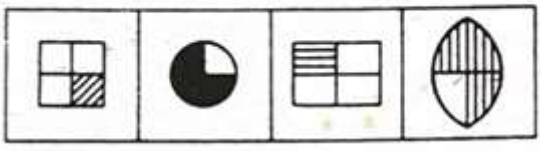
ANSWER FIGURES



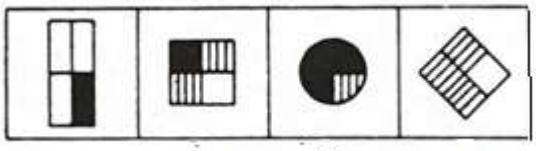
(a) (b) (c) (d)

99.

PROBLEM FIGURES

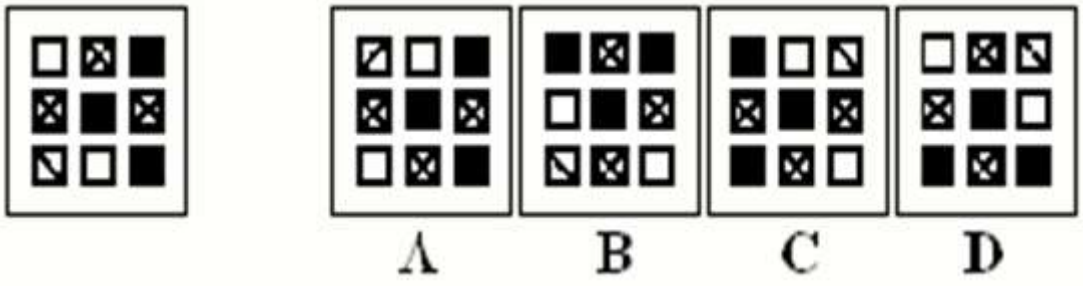


ANSWER FIGURES



(a) (b) (c) (d)

100. In the figures given below, one of the shapes (A –D) is identical to the first figure, but has been rotated. Which figure is identical to the first?



(a) A
(b) B
(c) C
(d) D

* * * * * * *